Interpretability

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Motivation

A model can be right for the wrong reason

(a) Husky classified as wolf (b) Explanation

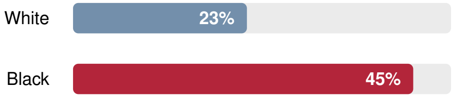
- A classifier told **huskies from wolves** almost perfectly
- An explanation revealed it keyed on the snow, not the animal

→ **High accuracy, useless reasoning**

Source: Ribeiro et al. 2016 [23]

When the reason is wrong, the stakes are high

COMPAS, flagged high-risk, but did not reoffend



False-positive rate, by group (ProPublica)

Biased scores

COMPAS flagged Black defendants who did not reoffend about twice as often.

Hidden shortcut

A pneumonia model learned asthma lowered risk, because those patients got faster care.

Leaky feature

An Amazon hiring model penalised the word "women's".

Source: ProPublica 2016; Caruana et al. 2015; Reuters 2018

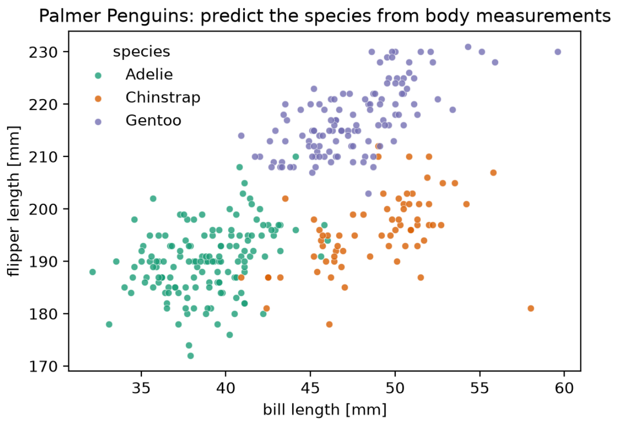
Why interpretability matters

- **Trust:** users are often unwilling to accept important decisions without understandable reasons
- **Learning:** a model can hold knowledge about the data that is valuable beyond the prediction itself
- **Debugging:** explanations help reveal shortcuts, leakage, or spurious correlations
- **Fairness & auditing:** interpretability can uncover harmful biases against particular groups
- **Safety:** in high-stakes settings, understanding model behaviour is part of responsible deployment

Source: [13], [38]

Our running example: the Palmer penguins

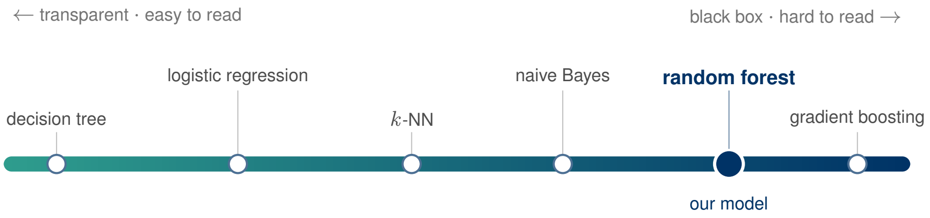
- Three species near Palmer Station: **Adelie**, **Chinstrap**, **Gentoo**
- Five measurements per bird: bill length and depth, flipper length, body mass, sex
- Task: predict the species from these measurements



Slides follow Christoph Molnar, Interpretable Machine Learning. Data: `palmerpenguins`.

Source: [13]

A random forest predicts the species accurately



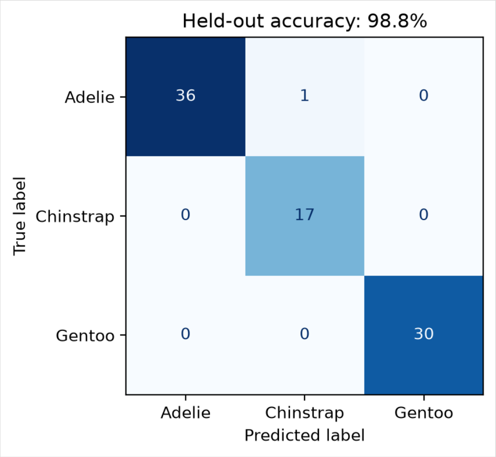
- You have met classifiers ranging from transparent to opaque

- Interpretability tends to fall as accuracy rises

- So we pick an accurate model, a **random forest**, and explain it afterwards

Source: [13]

Accuracy alone: 98.8 % on held-out birds

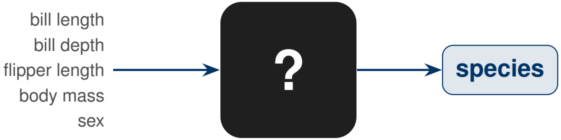


- Almost always right on birds it never saw
- Only one Adelie confused with a Chinstrap

→ **So, are we done?**

Source: [13]

A high score is not yet a reason to trust



a random forest, hundreds of trees, no single readable rule

- Which measurements does it really use?
- Did it take a shortcut, say sex or island, instead of biology?
- Would you trust it for a medical or credit decision?

→ **Accuracy tells us how often, never why**

Source: [13], [38], [144]

Interpretability

What do we mean by interpretability?

Doshi-Velez & Kim, 2017

"The ability to explain or to present in understandable terms to a human."

Miller, 2019

"The degree to which a human can understand the cause of a decision."

- No single formal definition, it is relative to the **audience** and the **task**

- Not directly measurable, we use **proxies** (sparsity, depth) or human studies

- Three evaluation levels: **application-**, **human-**, and **functionally-grounded**

Source: [38], [37]

What makes an explanation human-friendly

Good explanations follow the same patterns people use with each other (Miller, 2019).

Contrastive

Why this rather than another? People ask for differences.

Selective

A few strong reasons, not every detail.

Focus on the abnormal

Unusual values that flipped the prediction make the best reasons.

Social

Tailored to its audience and the question they actually asked.

Truthful & consistent

Holds on new cases and agrees with what the model really does.

Rule of thumb

Short, plausible, and answers the question actually asked.

Source: [13], [37]

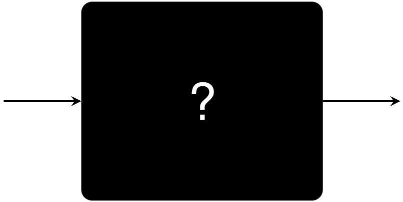
Two routes: interpretable model vs post-hoc explanation

Interpretable model

if flipper > 207
then Gentoo
$\beta_1 x_1 + \beta_2 x_2 + \ldots$

read the reasoning directly

Post-hoc explanation

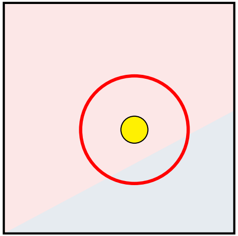


probe inputs, watch outputs, explain afterwards

- **Interpretable model:** simple enough to read on its own
- **Post-hoc:** keep the black box, explain its predictions afterwards
- The most accurate models are opaque, so post-hoc tools matter

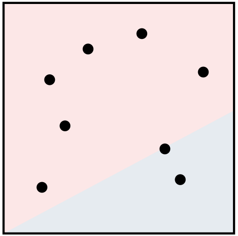
Source: [13]

A second axis: local vs global



Local

one prediction



Global

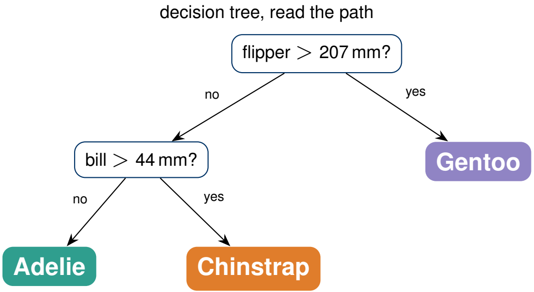
the whole model

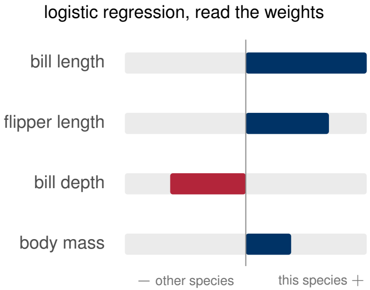
- **Local:** why this one prediction?
- **Global:** how does the model behave overall?

→ **We start local, explaining one bird at a time**

Source: [13]

Some models explain themselves





➔ **Our forest is neither shallow nor linear, so we need another way**

Source: [13]

A caution before we begin

Explanations can mislead

Unstable	Gameable	Not proof
Two runs can produce different explanations.	A model can be built to fool them.	An explanation is evidence, not proof.

→ **Always ask whether an explanation is stable and plausible before trusting it.**

Source: [13], [74]

Three ways to explain one prediction

Each method turns a prediction into a short list, **one number per feature**.

LIME *a local slope*

Fit one simple line near this bird, read its slopes.

What is the model doing right here?

Shapley *a fair share*

Give each feature its average contribution over all join orders.

What does each feature deserve?

SHAP *made practical*

The same fair share, estimated fast with ready-made plots.

How do we make it practical?

→ All three fit a readable stand-in near this one bird; they differ only in what counts as "simple" and "nearby".

Source: [23], [77], [80]

LIME

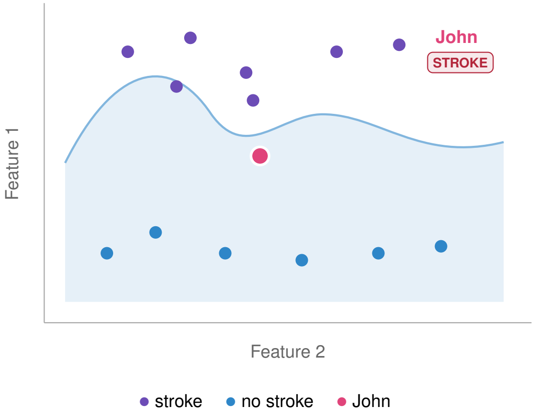
Local, interpretable, model-agnostic explanations

- **Local**, one prediction at a time
- **Interpretable**, a surrogate simple enough to read
- **Model-agnostic**, treats the model as a black box
- **Explanations**, as per-feature contributions

→ **A simple model, fit to be faithful only near one prediction.**

Source: [23], [13]

The idea: a simple model, valid nearby

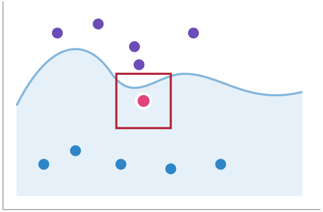


- A trained black box splits the space into regions, **stroke** and **no stroke**
- The whole boundary is **too tangled** to explain in one shot

→ **So explain just one prediction, John's, using only what happens nearby**

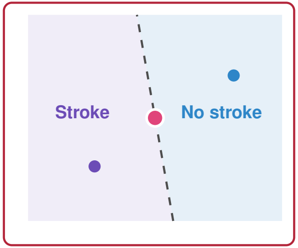
Source: [23], [13]

Zoom in: nearby, a straight line is enough



global, too complex

ZOOM IN



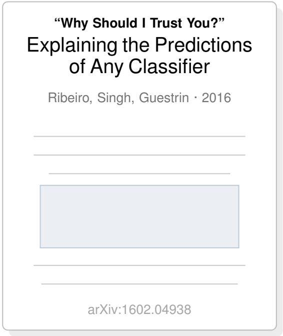
local, a straight line

→ **In this small neighbourhood one straight line separates the classes; that simple, readable model is LIME's explanation for John**

Source: [23], [13]

One recipe for any black box

- Works on **any** black-box model
- Model internals stay **hidden**, only inputs and outputs
- Works with many data types: **tables, text, images**



Source: [23]

The same idea, as one objective

Find the interpretable model g that best explains the black box $\hat{f}$ near the point $\mathbf{x}$.

$$\xi(\mathbf{x}) = \text{argmin}_{g \in G} \; \overbrace{L(\hat{f}, g, \pi_{\mathbf{x}})}^{\text{good local approximation}} + \overbrace{\Omega(g)}^{\text{stay simple}}$$

- $\hat{f}$: the complex black box

- g: a simple, interpretable model

- G: family of simple models, e.g. sparse linear

- $\pi_{\mathbf{x}}$: proximity, the local neighbourhood

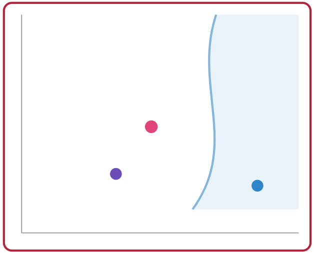
Source: [23], [13]

How to train the surrogate

$$\xi(\mathbf{x}) = \underset{g \in G}{\text{argmin}} \; \boxed{L(\hat{f}, g, \pi_{\mathbf{x}})} + \Omega(g)$$

- To minimise the boxed term we need data **near the instance**, and there is none yet

→ **Start from the one point we want to explain.**



Source: [23], [13]

How to train the surrogate

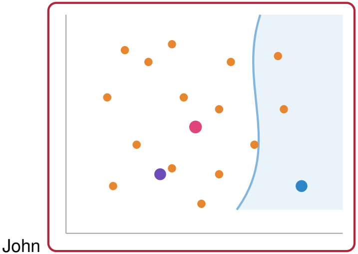
$$\xi(\mathbf{x}) = \underset{g \in G}{\operatorname{argmin}} \; \boxed{L(\hat{f}, g, \pi_{\mathbf{x}})} + \Omega(g)$$

- **Perturb** around it, then **label** each new point with $\hat{f}$

New dataset

labels: predictions of the black box $\hat{f}$

features: the perturbed points



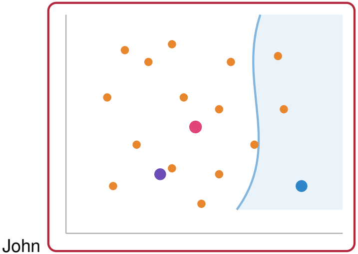
Source: [23], [13]

How to train the surrogate

- Fit g by **weighted least squares** on the new dataset

$$L = \sum_{z} \pi_{\mathbf{x}}(z) \left(\hat{f}(z) - g(z)\right)^2$$

- Near points (large $\pi_{\mathbf{x}}$) dominate the sum; far ones barely count



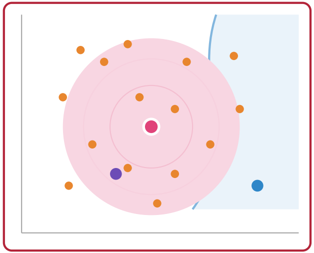
Source: [23], [13]

How to train the surrogate

$$L = \sum_z \boxed{\pi_{\mathbf{x}}(z)} \left(\hat{f}(z) - g(z)\right)^2$$

- The **proximity kernel** $\pi_{\mathbf{x}}$ turns distance into a weight (the pink glow)

→ **The kernel width sets how local the explanation is, a genuine choice.**



John

Source: [23], [13]

Neighbourhoods and kernels

"Nearby" is defined by a distance and a kernel, and that definition is a real choice.

$$\pi_{\mathbf{x}}(z) = \exp\left(-\frac{D(\mathbf{x}, z)^2}{\sigma^2}\right)$$

- D: a distance on standardised features (or on the interpretable z')

- Small width σ: very local, few effective points, **unstable**

- Large width σ: broader region, but **less faithful** to $\hat{f}$ near $\mathbf{x}$

→ There is no principled default width, so the neighbourhood is a genuine modelling choice.

Source: [23], [13]

The recipe, end to end

1. Pick the instance $\mathbf{x}$ → **2. Perturb** sample nearby → **3. Label** query $\hat{f}$ → **4. Fit** weight by $\pi_{\mathbf{x}}$, linear g → **5. Read** g's weights

$z' \in \{0, 1\}^d$

interpretable representation: which features (or superpixels) are on

$g(z') = \mathbf{w}^\top z'$

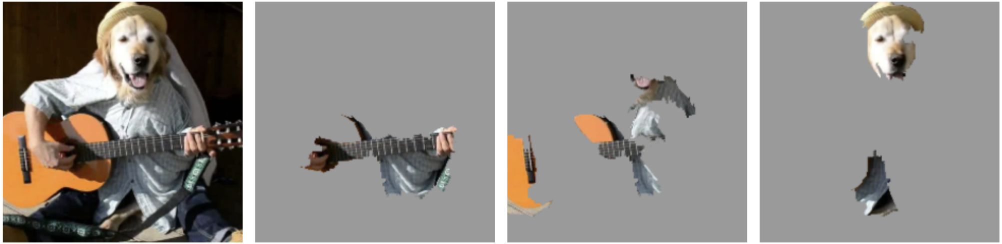
local linear surrogate: its weights are the explanation

$\Omega(g)$

keep g simple, only a few features so it stays readable

Source: [23], [13]

The same recipe explains images



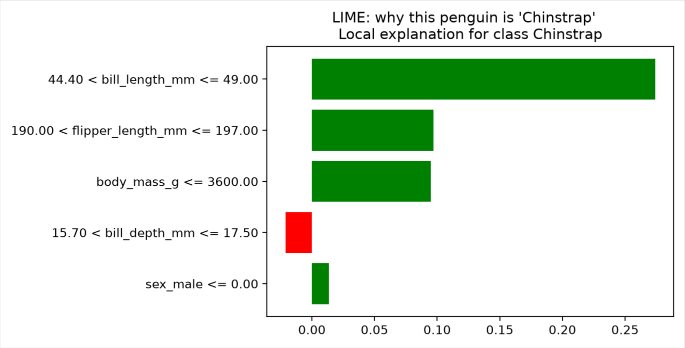
(a) Original Image (b) Explaining *Electric guitar* (c) Explaining *Acoustic guitar* (d) Explaining *Labrador*

- Group pixels into **superpixels**, then switch them on and off

- Each class highlights its own **responsible region**: the guitar for guitar, the dog's face for Labrador

Source: Ribeiro et al. 2016 [23]

Our penguin, explained by LIME

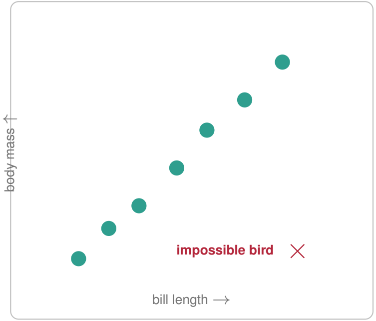


- **Green** pushes toward the predicted species
- **Red** pushes against it

→ Bill length matters most for this bird

Source: [23], [13]

A pitfall: impossible penguins



- LIME often perturbs each measurement **on its own**
- But bill length and body mass **move together**

Watch out

Independent perturbations can invent a bird in the empty space, one that cannot exist.

Source: [13]

What can go wrong?

Is the pipeline behind LIME deterministic?

No, the perturbations are **sampled at random**, so re-running LIME on the same instance can give **different explanations**.

- **Instability:** another random sample $\Rightarrow$ another story
- **Kernel width:** the neighbourhood is an arbitrary choice that changes the result
- **Correlated features:** independent perturbations invent impossible instances

→ **LIME can even be fooled: an adversarial model can hide biased behaviour from it.**

Source: [13], [74]

Strengths and limitations

Strengths

+ works on any model
+ intuitive and selective
+ tables, text, and images

Limitations

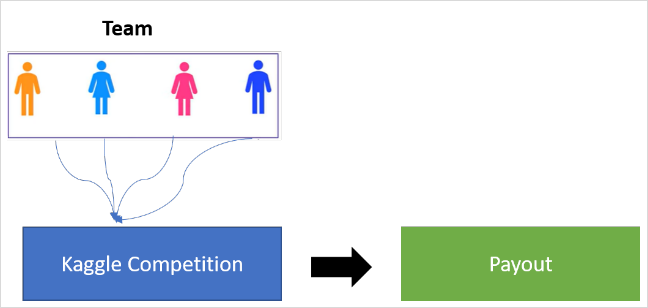
— the neighbourhood is an arbitrary choice
— results can be unstable
— no guarantee the split is fair

→ **Can we share a prediction in a way that is uniquely fair, with no arbitrary neighbourhood?**

Source: [13]

Shapley Values

The idea: a fair share of the prediction



- The **players** are the feature values
- The **game** is the prediction task for one instance
- The **payout** is the prediction minus a baseline prediction

→ If the features cooperate to reach the prediction, how should the payout be shared, and how do we even measure one feature's contribution?

Source: [13], [77]

Warm-up: feature contributions in a linear model

$$\hat{f}(\mathbf{x}) = \beta_0 + \beta_1 x_1 + \cdots + \beta_p x_p$$

Against the average, feature j's contribution to this prediction is simply

$$\phi_j(\mathbf{x}) = \beta_j x_j - \mathbb{E}[\beta_j X_j] = \beta_j \left(x_j - \mathbb{E}[X_j]\right)$$

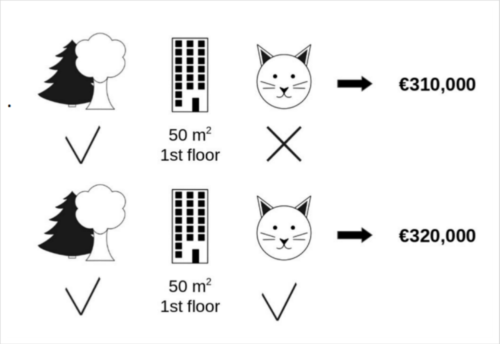
- Weight $\times$ centred feature value, read straight off the model

- They add up: $\sum_j \phi_j = \hat{f}(\mathbf{x}) - \mathbb{E}[\hat{f}(\mathbf{X})]$

→ **Clean and additive, but only because the model is linear. How do we generalise?**

Source: [13]

Marginal contributions

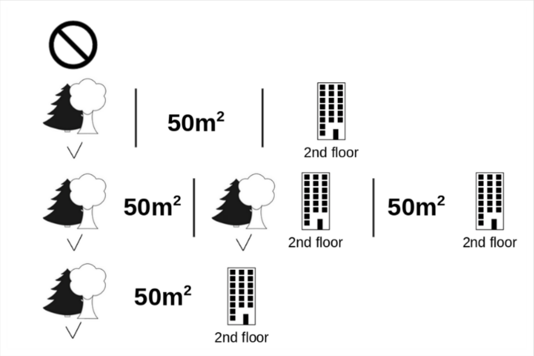


- Apartment price from **park**, **size**, and whether a **cat** is allowed
- Add **cat** to {park, size}: the price rises from **310k** to **320k**

→ **Marginal contribution of feature j to a coalition S: $v(S \cup \{j\}) - v(S)$**

Source: [13]

Which coalition? Average over all of them

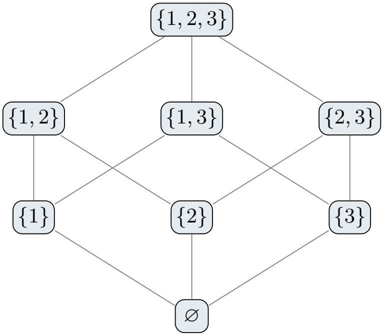


- The marginal contribution depends on which coalition S the feature joins
- So look at **every** coalition of the other features (park, size, floor)

→ **A feature's Shapley value is its average marginal contribution over all these coalitions**

Source: [13]

A fair average needs every coalition



- Averaging over orders means visiting **every subset** of features
- This small example: **3** features give **8** subsets
- Our penguin model uses **5** measurements, so $2^5 = $ **32** subsets

→ **The count doubles with every feature added**

Source: [13]

How to handle “missing” features

A coalition S fixes only some feature values. To query the model, the “absent” ones must be filled in.

present ($\mathbf{x}_S$, fixed)

absent

bill len	flipper	body mass	bill depth
39.1	**181**	?	?
		3.4k	18.1
		4.6k	15.2
		3.9k	19.0

sampled from background birds

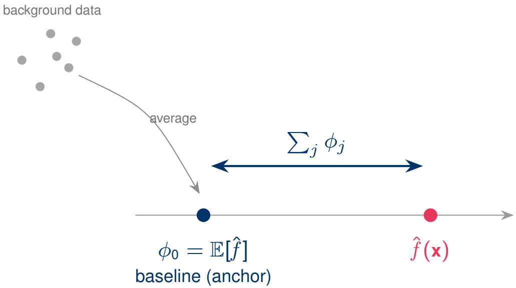
- Fix present features at $\mathbf{x}_S$; **sample** the absent ones from the data
- **Average** the resulting predictions to value the coalition

$$v(S) = \mathbb{E}\left[\hat{f}(\mathbf{X}) \mid \mathbf{X}_S = \mathbf{x}_S\right]$$

Empty: $v(\varnothing) = \mathbb{E}[\hat{f}]$ Full: $v(N) = \hat{f}(\mathbf{x})$

Source: [13], [80]

The background and the baseline



Background (reference data)

The set the absent features are drawn from; $v(S)$ averages the model over it. A different background changes the shares.

Baseline / base value ϕ_0 (anchor)

$\phi_0 = v(\varnothing) = \mathbb{E}[\hat{f}(\mathbf{X})]$, the average prediction. Every share is measured relative to this anchor.

→ **Efficiency: $\hat{f}(\mathbf{x}) = \phi_0 + \sum_j \phi_j$ ties the shares to the anchor.**

Source: [77], [80], [13]

The Shapley value, in closed form

The general rule: average the marginal contribution over every coalition, the linear warm-up was just the special case.

$$\phi_j = \sum_{S \subseteq N \setminus \{j\}} \overbrace{\frac{|S|!\,(p - |S| - 1)!}{p!}}^{\text{combinatorial weight}} \cdot \overbrace{\big(v(S \cup \{j\}) - v(S)\big)}^{\text{marginal contribution of } j}$$

- Average j's marginal contribution over every coalition S that excludes it

- The weight is the share of orderings; equivalently, average over all $p!$ orders

- For a linear model this reduces to the warm-up $\beta_j\big(x_j - \mathbb{E}[X_j]\big)$

Source: [77], [13]

What “fair” means: four axioms

Any sensible way to split a prediction should satisfy all four.

Efficiency

The shares add up to the whole gap: $\sum_j \phi_j = \hat{f}(\mathbf{x}) - \mathbb{E}[\hat{f}]$. Nothing lost, nothing invented.

Symmetry

Two features with the same marginal contribution in every coalition get the same share.

Null player

A feature that adds nothing to any coalition gets exactly zero.

Additivity

Combine (add) two models, and their shares add too.

Source: [77], [13]

Where the axioms live in the formula

$$\phi_j = \sum_{S \subseteq N \setminus \{j\}} \underbrace{\frac{|S|!\,(p-|S|-1)!}{p!}}_{\text{weight}} \cdot \underbrace{\big(v(S \cup \{j\}) - v(S)\big)}_{\text{contribution}}$$

- **Efficiency:** weights make shares sum to $v(N) - v(\varnothing)$
- **Null player:** zero contribution everywhere $\Rightarrow$ $\phi_j = 0$
- **Symmetry:** weight depends only on $|S|$; only contributions enter
- **Additivity:** ϕ is linear in v

Source: [77], [13]

From a formula to a proof

Theorem (Shapley, 1953)

One <u>and only one</u> attribution rule satisfies all four axioms at once: the average marginal contribution ϕ_j.

- Efficiency, Symmetry, Null player, Additivity $\Rightarrow$ exactly one rule

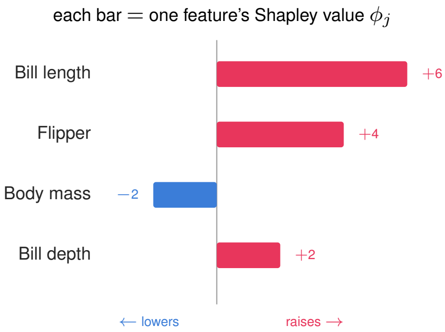
- Not <u>a</u> way to divide the prediction, but <u>the</u> fair way. A proof, not a trick

LIME cannot offer this

Its neighbourhood, kernel and penalty are free choices, so it has <u>none</u> of these four guarantees.

Source: [77], [13]

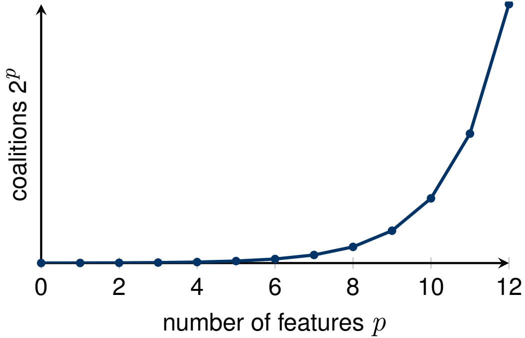
Reading a Shapley value in practice



- Start at the average prediction ϕ_0; each ϕ_j nudges it toward this bird
- Red raises the prediction, blue lowers it
- They sum exactly to the gap: $6 + 4 - 2 + 2 = 10 = \hat{f}(\mathbf{x}) - \mathbb{E}[\hat{f}]$
- An **attribution** under the model, <u>not</u> a causal effect

Source: [13], [77]

Exact Shapley values are too expensive



- Exact values need all 2^p coalitions
- Hopeless beyond a handful of features
- Instead, **estimate** by sampling random orders

→ **Can we make this fast, with ready-made tools?**

Source: [13], [78]

SHAP

SHAP: SHapley Additive exPlanations

SHapley Additive exPlanations

The same fair Shapley shares, turned into a practical, well-tooled framework (Lundberg & Lee, 2017): a modern interface to Shapley-style explanations.

Practical

fast estimators (KernelSHAP for any model, TreeSHAP for trees), not the 2^p sum

Unified

one quantity, local (one bird) and global (all birds, averaged)

Visual

a family of ready-made plots: waterfall, importance, dependence

Source: [80], [78], [13]

From Shapley values to additive explanations

$$g(z') = \phi_0 + \sum_{j=1}^{M} \phi_j \, z'_j, \qquad z' \in \{0, 1\}^M$$

For the instance itself all features are present ($z' = \mathbf{1}$), so $\hat{f}(\mathbf{x}) = \phi_0 + \sum_j \phi_j$: the efficiency equation.

- z': **simplified features**, present or absent (superpixels for images)

- ϕ_0 is the baseline, each ϕ_j a feature attribution

- LIME and SHAP are both additive; SHAP fixes ϕ_j to the Shapley values

Source: [80], [13]

SHAP's three properties

SHAP fixes the coefficients so the explanation obeys three properties (Lundberg & Lee, 2017).

Local accuracy

The explanation reproduces the prediction: $\hat{f}(\mathbf{x}) = \phi_0 + \sum_j \phi_j$, with $\phi_0 = \mathbb{E}[\hat{f}]$.

Missingness

An absent feature gets no attribution: $z'_j = 0 \Rightarrow \phi_j = 0$.

Consistency

If a feature becomes more influential, its SHAP value never decreases.

→ **These echo the Shapley fairness axioms, now stated for additive explanations.**

Source: [80], [13]

The Shapley kernel makes the fit exact

$$\pi_{\mathbf{x}}(z') = \frac{M - 1}{\binom{M}{|z'|} |z'| (M - |z'|)}$$

- KernelSHAP minimises $\sum_{z'} \pi_{\mathbf{x}}(z')(\hat{f}(h_{\mathbf{x}}(z')) - g(z'))^2$: weighted least squares, **no penalty**

- With this kernel, the solution equals the Shapley values

- LIME picks kernel and penalty heuristically; here the kernel is uniquely fixed

Source: [80], [13]

From exact Shapley to SHAP: what gets simplified

Exact Shapley

cost 2^p

$$\phi_j = \sum_{S \subseteq N \setminus \{j\}} w(S)\,\big(v(S \cup \{j\}) - v(S)\big)$$

KernelSHAP

one WLS solve

$$\phi = \operatorname*{argmin}_{\phi} \sum_{z' \in Z} \pi_{\mathbf{x}}(z')\,\big(\hat{f}(h_{\mathbf{x}}(z')) - g_{\phi}(z')\big)^2$$

~~every 2^p coalition~~

$\downarrow$ a sample Z, one weighted least squares

~~conditional $\mathbb{E}[\hat{f} \mid \mathbf{x}_S]$~~

$\downarrow$ average over a background set, features independent

~~combinatorial weights $w(S)$~~

$\downarrow$ the Shapley kernel $\pi_{\mathbf{x}}$, exact as $Z \to$ all

→ Same target, the fair shares. SHAP trades an exact exponential sum for a weighted regression on a sample, plus a feature-independence assumption.

Source: [80], [166], [13]

How KernelSHAP estimates the shares

→ It is LIME's recipe (perturb, predict, fit) with two twists: the samples are feature coalitions, and a special weighting returns the exact Shapley values

A coalition = which features are present

Bill length	Flipper	Body mass	Bill depth
1	**0**	**1**	**0**

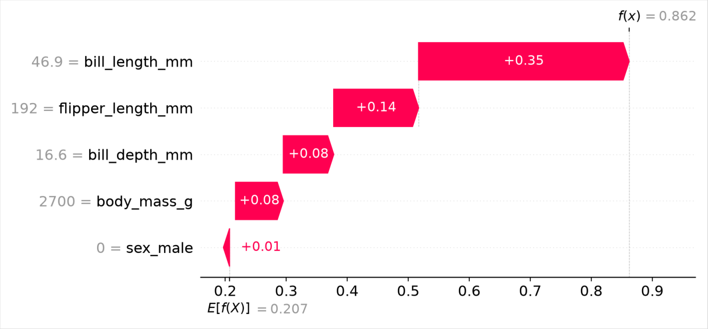
$$z' = (1, 0, 1, 0) \in \{0, 1\}^M$$

1 present, the bird's value **0** absent, from a background bird

- **Sample** many coalitions z'
- **Fill** absent features from a background bird
- **Query** the model $\hat{f}$ on each; a **weighted fit** gives the shares ϕ_j

Source: [80], [13]

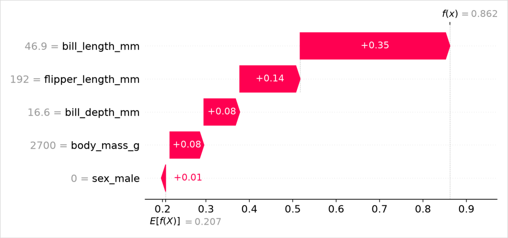
Reading one prediction



- Start at the **average** prediction, end at this bird's prediction
- **Red** pushes up, **blue** pushes down
- Each bar is one feature's fair share

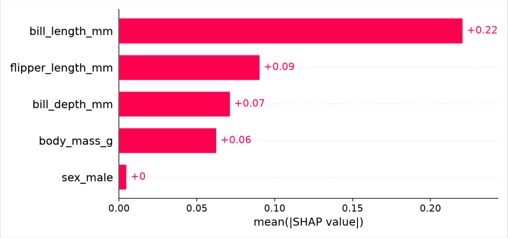
Source: Molnar 2023 [13]

From one bird to the whole model



one bird, local shares

→

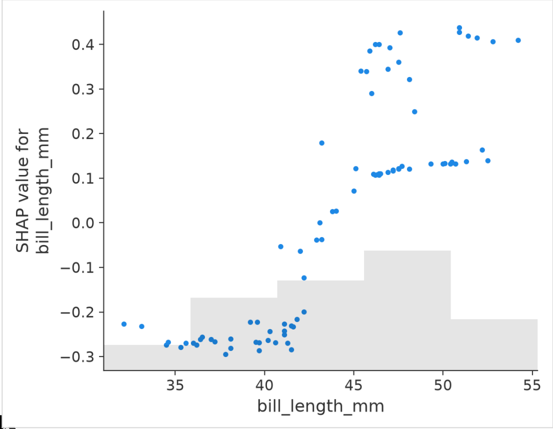


all birds, global importance

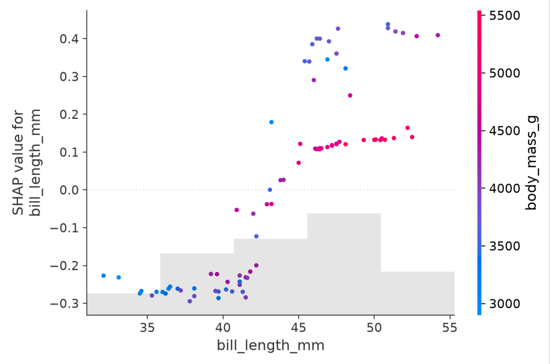
→ Average the shares over all birds: one framework, both local and global.

Source: Molnar 2023 [13]

Effects and interactions



how the share changes with a feature's value



colour reveals an interaction with another feature

Source: [13], [165]

Strengths and limitations

Strengths

+ a solid game-theoretic footing
+ one framework, local and global
+ fast and exact for trees

Limitations

— the general estimator can be slow
— the chosen background matters
— still not a causal statement

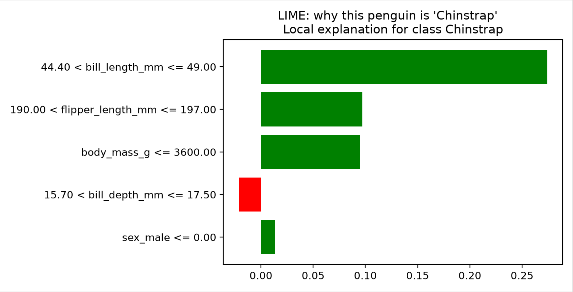
→ **SHAP is the practical Shapley framework, but the same cautions remain**

Source: [13], [74]

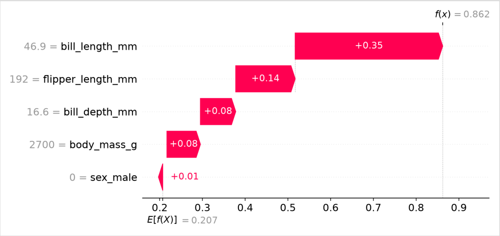
Comparison

Our penguin, explained two ways

LIME



SHAP



→ **Both rank bill length first, but LIME reports a local slope while SHAP splits the prediction against a baseline**

Source: [23], [80], [13]

Same bird, two different questions

Both charts rank **bill length** first, yet their numbers answer different questions.

LIME asks: what is happening right here?

Each bar is a **local surrogate weight**: how strongly a measurement shifts the score in this bird's neighbourhood. Green pushes toward the predicted species, red against. It is a local slope, not a total.

SHAP asks: what does each feature fairly deserve?

Each bar is a **fair share**; starting from the baseline ϕ_0 (the average prediction) they add up exactly to this bird's prediction: $\hat{f}(\mathbf{x}) = \phi_0 + \sum_j \phi_j$.

→ A local slope (LIME) is not the same as a fair split (SHAP), even when they agree on the top feature.

Source: [23], [80], [13]

Pros and cons at a glance

Method	Good for	Watch out for
LIME	quick, intuitive local pictures on any model	arbitrary neighbourhood, unstable, no fairness guarantee
Shapley	a uniquely fair split with a clear theory	exponential cost, sensitive to the background
SHAP	fast Shapley in practice, good plots, tree support	inherits the Shapley cautions

→ **LIME for quick intuition, Shapley for theory, SHAP for a practical fair split**

Source: [13], [23], [77], [80]

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